

I fell in love with teaching early in my undergraduate years, long before I held a formal teaching appointment. At Obafemi Awolowo University, several classmates found *Mechanics of Machines I* difficult, even though the physical ideas behind the equations felt intuitive to me. I began organizing informal sessions in which I broke difficult problems into simpler steps, connected equations to the motion of real mechanical systems, and asked classmates to explain the reasoning back to me. What stayed with me was the joy I saw in my classmates when a concept that had seemed inaccessible became understandable. That experience made teaching a central part of the academic career I wanted. It also strengthened my desire to develop deeper technical expertise so that I could teach, mentor, and contribute new knowledge, which became one of my motivations for pursuing a doctoral study.

That interest grew into teaching experiences, including teaching mathematics, physics, science, and advanced-level mathematics in various institutions across my home country, as well as volunteering as a Graduate Record Examination (GRE) instructor for prospective graduate students, and later teaching Industrial Engineering students at Arizona State University. My ASU instructional experience includes IEE 380 *Probability and Statistics for Engineering Problem Solving*, IEE 470 *Quality Control*, and IEE 570 *Advanced Quality Control*. Across classroom teaching and GRE instruction, more than 300 learners have benefited from my teaching and educational support. These experiences have shaped three connected principles that guide my teaching. I aim to (1) make learning purposeful and active, (2) make reasoning and expectations transparent, and (3) use authentic engineering problems to develop critical thinking and independence.

1. Make learning purposeful and active. Engineering courses can quickly become collections of formulas, algorithms, and software commands if students do not understand what those tools make possible. I therefore begin by introducing the physical or operational problem before presenting the mathematical machinery. I ask three questions. What is being observed? What is uncertain? And what decision must be made? In probability and statistics, this shifts attention from simply selecting a formula to understanding what variability means for an engineering conclusion. In quality control, a control chart statistic becomes more meaningful when students see it as evidence for deciding whether a process has changed and what action should follow.

Purpose alone, however, is not enough to sustain engagement. Students now learn in an environment where social media and online content constantly compete for attention, so I strive to make class time participatory rather than relying on long, uninterrupted exposition. I intersperse explanations with questions, short problem-solving activities, interpretation of plots or process signals, and opportunities for students to explain why a method or assumption is appropriate. These moments of active participation also serve as real-time formative assessment because they show me where understanding is breaking down before confusion accumulates. My goal is to create enough opportunities for participation that students who are hesitant to speak first, or who enter with different levels of preparation, are not allowed to quietly fall behind.

2. Make reasoning visible and expectations transparent. Students become independent problem solvers when they can see how an expert approaches an unfamiliar problem and understand what strong work requires. I strive to organize difficult engineering problems around a repeatable sequence. Students first define the question, identify what is known and unknown, recognize sources of variability or uncertainty, select an appropriate model, state its assumptions, compute or simulate, check the result, and interpret it in the original context. Early assignments make these steps explicit. As students develop confidence, I remove more of the scaffolding so that they must decide which tools, assumptions, and diagnostic checks are necessary. This is particularly important in computational courses because a correct Python or software output is not necessarily evidence of correct engineering reasoning.

I use the same philosophy in assessment. A numerical answer matters, but so do model choice, assumptions, units, diagnostics, interpretation, reproducibility, and communication. This is evident in the Summer 2026 IEE 570 course that I taught. The term project required each team to explain an advanced monitoring methodology, demonstrate it through a manageable example, connect it to a real application, and discuss limitations and future research. I also

invited students to share drafts before submission and provided specific comments that they could use to revise their work. Transparent expectations and iterative feedback help students understand what successful work looks like, why a revision matters, and how to improve before the learning opportunity has passed.

3. Build critical thinking through authentic engineering problems. My goal is for students to leave a course able to do more than reproduce a method they have already seen. They should be able to decide whether a method is appropriate, defend assumptions, recognize limitations, and communicate a recommendation when the problem is incomplete. Authentic projects are especially valuable because they generate questions that do not arise when every quantity and method is specified in advance.

One exchange from Summer 2026 IEE 570 captures this process. A team studying Hidden Markov Models for change point detection contacted me because they were unsure what it meant to demonstrate the methodology through a simple example and what should be discussed as future work. Rather than give them a completed solution, I asked them to select or simulate data, apply the method themselves, generate their own results, and think about how the method could be improved. Their final work connected the method to a manufacturing maintenance problem and earned 92%. Across all ten teams in the course, project scores ranged from 89% to 98%, with eight teams scoring at least 92%. Student feedback reinforced the value of this approach. Unsolicited messages thanked me for a “great semester” and described my feedback and advice as “greatly beneficial.” For me, these outcomes illustrate the purpose of project-based learning. A project should not merely test what students know. It should create questions that require them to think more deeply.

Mentoring experience

My mentoring philosophy extends the same progression from guided reasoning to intellectual ownership. I currently jointly mentor a four-student ASU final year capstone project team, with students from backgrounds spanning Industrial Engineering and Engineering Management, on an image-based 3D reconstruction project for additive manufacturing quality assurance. The team began the two-semester project in Spring 2026 and will complete its second phase in Fall 2026. During the spring semester, I helped translate the broader research problem into objectives appropriate for a capstone project, shared foundational literature and technical materials, discussed the project in depth with the team, helped establish weekly progress expectations, reviewed their work, completed sponsor evaluations, and supported preparation for their capstone presentation. The project requires students to connect imaging hardware, calibration, photogrammetry, photometric stereo, reconstruction, and engineering quality decisions. Working with this interdisciplinary team has strengthened my ability to guide students with different technical backgrounds toward a common engineering objective. It has also prepared me to contribute to senior design, capstone, and other project-based courses.

I currently mentor a recent Mechanical Engineering graduate and a first-year Industrial Engineering Ph.D. student on the same broader 3D reconstruction research program. I use explicit short-term goals, regular research discussions, reproducible workflows, and iterative feedback to help students move from implementing individual tasks toward making and defending methodological choices. The objective is not simply to make a computational pipeline run, but to develop the habits needed to determine whether a result is physically meaningful and sufficiently validated. I have also mentored an international undergraduate researcher through ASU’s Summer Research Initiative on active learning research and worked with my advisor to mentor two master’s students on generative CAD modeling from hand-drawn sketches and inverse modeling for microstructural characterization using eddy current field data, respectively. Both master’s projects contributed to their theses. These experiences have taught me to scale the difficulty of a research problem to a student’s current preparation while preserving a clear path toward independence.

Mentoring also includes helping students see pathways into graduate education. From October 2020 through July 2021, I volunteered as a GRE instructor, teaching quantitative reasoning, verbal reasoning, analytical writing, test-taking strategy, and structured problem-solving to prospective graduate students. In 2021, EducationUSA Abuja invited me to share GRE and TOEFL/Duolingo preparation strategies, and I returned as a resource person in 2024. I continue to share preparation materials, practical advice, and graduate application resources because access to clear information

and constructive guidance can materially change which opportunities students view as attainable.

Teaching plans and continuous improvement

My background in industrial engineering, statistics, scientific machine learning, uncertainty quantification, and manufacturing prepares me to teach undergraduate and graduate courses in probability and statistics, quality and reliability engineering, design of experiments, data analytics, engineering simulation, machine learning, manufacturing systems, and uncertainty quantification. My capstone mentoring has also prepared me to contribute to senior design and project based engineering courses. I am particularly interested in developing an undergraduate *Manufacturing Data Analytics* course in which students work with vibration, temperature, image, or other process data, and a graduate *Physics Based Machine Learning for Engineers* course in which students build and validate surrogate, operator learning, or physics discovery models for physical systems. These courses would connect directly to my research program. My research asks how to infer hidden physical quantities from imperfect observations, discover missing governing structure, quantify uncertainty, and make computational models useful for repeated engineering decisions. I want students to confront the same questions in the classroom. What is actually observed? What assumptions connect the data to the physical system? When should a model be trusted, updated, or rejected? Manufacturing, materials, thermal, and semiconductor systems provide natural settings in which statistical reasoning, computation, and physics must work together.

I view teaching itself as an iterative engineering process. Student questions, participation, project quality, concept level performance, and structured feedback provide evidence about where examples, pacing, scaffolding, or assessments need to change. I use those signals to refine the course rather than assume that one successful explanation will work for every group of students. Ultimately, I want students to leave my classroom able to explain not only how an answer was obtained, but why it is reasonable, what assumptions support it, what uncertainty remains, and what engineering decision it enables. The same satisfaction that first drew me to teaching as an undergraduate still motivates me. I want students to move from seeing a difficult concept as a barrier to seeing it as a tool they can use independently.